HRL for Dynamic Affine Maximizer Auctions: Strict Incentive Compatibility via Hierarchical Revenue Optimization

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April 16, 2026

Abstract

We present *HRL-AMA*, a framework for dynamic combinatorial auction design that simultaneously achieves exact dominant-strategy incentive compatibility (DSIC) and high cumulative revenue over a multi-round horizon. Existing approaches face a fundamental trilemma: neural-network mechanisms such as RegretNet learn high-revenue auctions but can only guarantee approximate DSIC (ε-IC, $\varepsilon > 0$); classical mechanisms such as VCG are exactly DSIC but static; and direct RL over all possible mechanisms incurs exponentially large action spaces. HRL-AMA resolves this trilemma by (i) fixing the mechanism structure to be an Affine Maximizer Auction (AMA), which guarantees exact DSIC for any parameter values satisfying $w_i > 0$, and (ii) employing a hierarchical RL agent to dynamically tune the AMA parameters—*bidder boosts* w_t and *item-assignment weights* ϕ_t—across rounds. A high-level controller (PPO) observes macro market states and outputs a strategic goal vector every K rounds; a low-level worker (SAC) translates the goal into concrete AMA parameters each round; a deterministic AMA solver clears the auction with provable DSIC and individual rationality (IR). We prove that per-round DSIC holds for *any* RL policy (Theorem 8.1), with per-round IC regret identically zero, contrasting sharply with the $\varepsilon > 0$ regret of differentiable mechanism design approaches. Empirical benchmarks on four valuation distributions (Uniform, Lognormal, LLG, Non-stationary) with $n = 3$ bidders, $m \in \{2, 3\}$ items, and $T = 30$ rounds confirm that dynamically tuned AMA parameters outperform static mechanisms, with the non-stationary distribution motivating the hierarchical architecture.

1 Introduction

Combinatorial auctions—in which n bidders compete for bundles of m heterogeneous items—are central to resource allocation in spectrum licensing, cloud computing, and online advertising. A practically deployable auction mechanism must satisfy three properties simultaneously: (i) *exact dominant-strategy incentive compatibility* (DSIC), so that truthful reporting is each bidder's undominated strategy regardless of others' behaviour; (ii) *individual rationality* (IR), so that participation is always weakly preferred to abstention; and (iii) *revenue optimisation*, so that the auctioneer captures high total payments over a repeated interaction horizon.

Achieving all three simultaneously has proved elusive. The Vickrey–Clarke–Groves (VCG) mechanism [Vickrey, 1961, Clarke, 1971, Groves, 1973] provides exact DSIC and IR by construction, but its parameters are fixed and revenue-suboptimal in general. Differentiable mechanism design approaches, notably RegretNet [Dütting et al., 2019], learn high-revenue mechanisms from data but can only enforce approximate DSIC (ε-IC), leaving a non-zero incentive for strategic misreporting. Applying reinforcement learning (RL) directly to the mechanism space yields adaptive policies but faces action spaces of size $(n+1)^m$—exponential in the number of items—and provides no IC guarantee.

Key insight. Incentive compatibility in an Affine Maximizer Auction (AMA) is a *structural property* of the mechanism, independent of the specific parameter values. Concretely, for any bidder boosts $w_i > 0$ and any item-assignment weights $\phi_{ij} \in \mathbb{R}$, the AMA allocation and payment rules guarantee exact DSIC and IR. This observation admits a clean architectural separation: IC is enforced by fixing the mechanism class to AMA, while revenue optimisation is entirely delegated to an RL agent that tunes AMA parameters over an unconstrained space—without any IC penalty term in its objective.

HRL-AMA. We propose HRL-AMA, which operationalises this separation via a two-level hierarchical RL architecture. The *high-level controller* observes evolving market conditions—bidder budget depletion, time progress, and historical bidding statistics—and outputs, every K rounds, a goal vector encoding macro pricing strategy. The *low-level worker* translates the current goal and fine-grained state into concrete AMA parameters (w_t, ϕ_t) each round. A deterministic *AMA solver* then clears the combinatorial auction with provably exact DSIC and IR.

Contributions. This paper makes the following contributions.

1. We propose the **HRL-AMA framework**, the first approach to couple exact-DSIC AMA mechanisms with hierarchical RL for dynamic combinatorial auction design.
2. We establish the **IC independence property** of AMA: DSIC and IR hold for any parameter values (w, ϕ) with $w_i > 0$, enabling an unconstrained RL objective without IC-penalty terms.
3. We prove **Theorem 8.1** (Dynamic DSIC): for any RL policy and any round, if the output weights satisfy $w_{i,t} > 0$, then per-round DSIC holds with IC regret identically zero.
4. We introduce a **linear item-assignment weight parameterisation** $\phi \in \mathbb{R}^{n\times m}$ that encodes all $(n+1)^m$ allocation boosts, reducing the RL action dimension from exponential to $n(1+m)$.
5. We provide a **fully verified reference implementation**: a Gymnasium combinatorial auction environment, three baseline mechanisms (VCG, Static AMA, Flat DRL), and an automated test suite of 128 tests confirming DSIC and IR across all experimental conditions.

Paper outline. Section 2 reviews related work. Section 3 formalises notation and the auction setting. Section 4 defines the AMA and proves its static DSIC and IR guarantees. Section 5 formulates the multi-round auction as an MDP. Section 6 presents the hierarchical RL architecture. Section 7 details the item-assignment weight parameterisation. Section 8 states and proves the main dynamic DSIC theorem. Section 9 reports empirical results. Section 10 discusses limitations and directions for future work. Section 11 concludes.

2 Background and Related Work

Combinatorial mechanism design. Roberts [Roberts, 1979] characterised all DSIC social-choice functions satisfying a mild range condition as affine maximisers, establishing the AMA family as the canonical class of exactly incentive-compatible mechanisms. Likhodedov and Sandholm [Likhodedov and Sandholm, 2004] showed that AMA parameters can be tuned for revenue under fixed distributional assumptions via offline optimisation. Our contribution extends this to the adaptive setting, in which parameters are adjusted round-by-round based on observed market dynamics.

Differentiable mechanism design. Dütting et al. [2019] introduced RegretNet, which parameterises allocation and payment rules as neural networks and enforces approximate DSIC via an augmented Lagrangian penalty, achieving ε-IC with $\varepsilon > 0$. Subsequent work [Feng et al., 2018] extends this framework to budget-constrained settings. These approaches trade exact IC for flexibility; the AMA structure eliminates this trade-off by guaranteeing exact DSIC unconditionally.

RL for dynamic auctions. Applying RL to adapt mechanism parameters over time has been studied in simpler single-item and posted-price settings. Lubin et al. [2022] surveys deep learning approaches to combinatorial auctions. To our knowledge, HRL-AMA is the first framework that couples RL-based parameter optimisation with a provably DSIC combinatorial auction mechanism.

Hierarchical reinforcement learning. The Options framework of Sutton et al. [1999] provides the theoretical foundation for two-timescale decision-making. We adopt this framework to decompose the auction optimisation problem: the high-level controller sets macro pricing strategy on a K-round timescale, while the low-level worker fine-tunes AMA parameters at the per-round timescale.

3 Problem Setup

3.1 Notation

We consider a combinatorial auction with n bidders indexed by $i \in [n] := \{1, \ldots, n\}$ and m items indexed by $j \in [m]$. Table 1 summarises the notation used throughout.

3.2 Feasible Allocations and Additive Valuations

An allocation $a = (S_1, \ldots, S_n)$ assigns each item to at most one bidder:

$$\mathcal{A} := \left\{ (S_1, \ldots, S_n) : S_i \subseteq [m],\ S_i \cap S_j = \emptyset \ \forall i \neq j \right\}. \tag{1}$$

Items may be left unallocated. We restrict to *additive* valuations: $v_i(S) = \sum_{j \in S} v_i[j]$ with $v_i[j] \geq 0$.

Table 1: Notation summary.

Symbol	**Meaning**	**Notes**
n, m, T	Number of bidders, items, rounds	
$v_i : 2^{[m]} \to \mathbb{R}_{\geq 0}$	True valuation of bidder i	private; $v_i(\emptyset) = 0$
b_i	Reported bid of bidder i	may differ from v_i
$\mathcal{A}$	Feasible allocation space	$\lvert\mathcal{A}\rvert = (n+1)^m$
S_i^a	Bundle assigned to bidder i under a	
$w_i > 0$	Bidder boost for bidder i	RL-tuned; must be positive
$\phi_{ij} \in \mathbb{R}$	Item-assignment weight	RL-tuned; unconstrained
$\lambda_a(\phi)$	Allocation boost derived from ϕ	linear; see (2)
p_i	Payment of bidder i	
B_i	Remaining budget of bidder i	decreases by p_i each round
u_i	Utility of bidder i: $v_i(S_i^{a^*}) - p_i$	IR requires $u_i \geq 0$

3.3 Valuation Distributions

We evaluate on four distributions spanning different structural challenges: uniform additive $v_{ij} \sim U(0,1)$; log-normal $v_{ij} \sim \mathrm{LogN}(0, 0.5)$ (heavy tail); the Local–Local–Global (LLG) model with two local bidders valuing single items and one global bidder valuing bundles (complementarity); and a non-stationary variant in which the scale shifts from $U(0,1)$ to $U(0,3)$ at round $T/2$ (distribution shift).

4 The Affine Maximizer Auction

4.1 Parameters and Mechanism Definition

Definition 4.1 (AMA Parameters)**.** The AMA is parameterised by two objects:

- *Bidder boosts* $w = (w_1, \ldots, w_n) \in \mathbb{R}^n_{>0}$. Setting $w_i > 1$ amplifies bidder i's contribution to the allocation objective, making the mechanism more likely to assign items to that bidder and reducing their payment via the $1/w_i$ scaling.
- *Item-assignment weights* $\phi \in \mathbb{R}^{n \times m}$. The entry ϕ_{ij} encodes the mechanism's intrinsic preference for assigning item j to bidder i, independently of submitted bids. Positive values encourage the assignment; negative values discourage it.

The item-assignment weights induce a scalar allocation boost via a linear inner product:

$$\lambda_a(\phi) \;:=\; \sum_{i=1}^{n} \sum_{j=1}^{m} \phi_{ij}\, \mathbf{1}[j \in S_i^a] \;=\; \langle \phi,\, \Psi(a) \rangle, \qquad [\Psi(a)]_{ij} = \mathbf{1}[j \in S_i^a]\,, \tag{2}$$

where $\Psi(a) \in \{0,1\}^{n \times m}$ is the binary assignment indicator matrix of allocation a.

Definition 4.2 (Affine Maximizer Auction)**.** Given parameters (w, ϕ), the AMA mechanism (a^*, p) is:

Allocation rule:

$$a^*(\mathbf{b};\, w, \phi) \;=\; \arg\max_{a\in\mathcal{A}} \Big[\sum_{i=1}^{n} w_i\, b_i(S_i^a) + \lambda_a(\phi) \Big]. \tag{3}$$

Payment rule (VCG-style):

$$p_i(\mathbf{b};\, w, \phi) \;=\; \frac{1}{w_i} \Big[\max_{a\in\mathcal{A}} W_{-i}(a) \;-\; W_{-i}(a^*) \Big], \tag{4}$$

where the *external welfare* excluding bidder i is

$$W_{-i}(a) \;:=\; \sum_{j\neq i} w_j\, b_j(S_j^a) + \lambda_a(\phi). \tag{5}$$

Remark 4.3 (Relation to VCG). Setting $w_i = 1$ for all i and $\phi = \mathbf{0}$ recovers the standard VCG mechanism. HRL-AMA strictly generalises VCG: the RL agent learns when and how to depart from equal-weight, zero-boost VCG in order to extract additional revenue.

4.2 Exact DSIC and IR: Structural Guarantees

Theorem 4.4 (Static DSIC). *For any $w \in \mathbb{R}^n_{>0}$ and any $\phi \in \mathbb{R}^{n\times m}$, the AMA mechanism satisfies dominant-strategy incentive compatibility: truthful reporting $b_i = v_i$ is weakly dominant for every bidder i, for all reports of others $\mathbf{b}_{-i}$.*

Proof. Fix $\mathbf{b}_{-i}$ arbitrarily; $W_{-i}(\cdot)$ is then a fixed function of allocations. Let $a^* = a^*(v_i, \mathbf{b}_{-i})$ and $a' = a^*(b_i', \mathbf{b}_{-i})$. Multiplying both utilities by $w_i > 0$:

$$\begin{aligned} w_i\, u_i^{\text{truth}} &= w_i\, v_i(S_i^{a^*}) + W_{-i}(a^*) - \max_a W_{-i}(a), \\ w_i\, u_i^{\text{dev}} &= w_i\, v_i(S_i^{a'}) + W_{-i}(a') - \max_a W_{-i}(a). \end{aligned}$$

Since a^* maximises $w_i\, v_i(S_i^a) + W_{-i}(a)$ over all $a \in \mathcal{A}$ (including a'), the first expression is no less than the second. Dividing by $w_i > 0$ yields $u_i^{\text{truth}} \geq u_i^{\text{dev}}$. $\square$

Theorem 4.5 (Individual Rationality). *For any $w \in \mathbb{R}^n_{>0}$, $\phi \in \mathbb{R}^{n\times m}$, and $\mathbf{b}_{-i} \geq 0$, the AMA satisfies ex-post individual rationality under truthful reporting: $u_i = v_i(S_i^{a^*}) - p_i \geq 0$.*

Proof. Let $a_0 \in \arg\max_a W_{-i}(a)$. By optimality of a^* and $v_i \geq 0$:

$$w_i\, v_i(S_i^{a^*}) + W_{-i}(a^*) \;\geq\; w_i\, v_i(S_i^{a_0}) + W_{-i}(a_0) \;\geq\; 0 + \max_a W_{-i}(a).$$

Therefore $w_i\, u_i = [w_i\, v_i(S_i^{a^*}) + W_{-i}(a^*)] - \max_a W_{-i}(a) \geq 0$, and $w_i > 0$ gives $u_i \geq 0$. $\square$

Theorems 4.4 and 4.5 are stated for arbitrary (w, ϕ) and hold without any condition on the specific parameter values beyond $w_i > 0$. This unconditional nature is what allows the RL agent in HRL-AMA to optimise revenue over an entirely unconstrained parameter space, with no IC penalty term in its objective.

Proposition 4.6 (IC Independence Property). *In the AMA, exact DSIC and IR are structural properties of the mechanism form, not of the parameter values. Any RL policy that maintains $w_{i,t} > 0$ for all i and t produces an exactly DSIC and IR mechanism at every round, regardless of the revenue objective it optimises.*

5 Multi-Round Auction as an MDP

5.1 Environment

The auction runs for T rounds, indexed $t = 1, \ldots, T$. At each round, bidder i draws a private valuation $v_{i,t} \sim \mathcal{F}_{i,t}$ and maintains a budget updated as $B_{i,t+1} = B_{i,t} - p_{i,t}$. Initial budgets $B_{i,0} \sim U(5, 15)$.

Budget exhaustion. If $B_{i,t} \leq 10^{-6}$, bidder i is removed from round t entirely: their bids are set to zero and they do not participate in the AMA. This prevents unenforceable payments that would arise when $\lambda_a(\phi) > 0$ might otherwise assign items to a zero-budget bidder.

5.2 State, Action, and Reward

State vector. The RL agent observes

$$s_t = \Big(t/T, \ \hat{\mu}_t \in \mathbb{R}^{n \times m}, \ \hat{\sigma}_t \in \mathbb{R}^{n \times m}, \ \bar{B}_t \in \mathbb{R}^n, \ \Delta r_t \in \mathbb{R}^K\Big), \tag{6}$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are running bid mean and standard deviation, $\bar{B}_t = B_t/B_0$ are normalised remaining budgets, and Δr_t is the vector of the K most recent per-round revenues. All components are computed from rounds $1, \ldots, t-1$ only, so the state s_t is independent of round-t bids—a condition essential for Theorem 8.1. **State dimension:** $1 + 2nm + n + K$ (equal to 27 for $n = 3$, $m = 3$, $K = 5$).

Action vector.

$$\theta_t = (w_{\mathrm{raw},t} \in \mathbb{R}^n, \ \phi_t \in \mathbb{R}^{n \times m}), \qquad w_{i,t} = \mathrm{softplus}(w_{\mathrm{raw},i,t}) = \log\big(1 + e^{w_{\mathrm{raw},i,t}}\big). \tag{7}$$

The softplus transformation guarantees $w_{i,t} > 0$ for all inputs, unconditionally satisfying the requirement of Proposition 4.6. **Action dimension:** $n + nm = n(1 + m)$ (equal to 12 for $n = m = 3$).

Reward and objective. The per-round reward is the total payment collected: $r_t = \sum_{i=1}^{n} p_{i,t}$. The RL objective is $\max_\pi \mathbb{E}_{\mathcal{F},\pi}[\sum_{t=1}^{T} r_t]$, where π maps states to AMA parameters.

5.3 Per-Round Timeline

The ordering of events within each round is critical for the dynamic DSIC result.

6 Hierarchical RL Architecture

6.1 Motivation for the Hierarchical Decomposition

A single-level RL agent outputting (w_t, ϕ_t) at every round faces three structural difficulties. First, the agent reacts to per-round bid noise and may fail to account for long-horizon market trends. Second, credit assignment over a T-step horizon is difficult, as early parameter choices receive attenuated reward signals. Third, a flat policy has no natural representation of macro pricing strategies such as budget-preserving behaviour in early rounds versus aggressive extraction near the end of the episode. Section 9 confirms empirically that a flat DRL agent (Flat DRL baseline) fails to adapt when the valuation distribution shifts mid-episode, providing direct motivation for the hierarchical architecture.

Step 1. RL agent observes s_t; outputs $(w_{\text{raw},t}, \phi_t)$ *Parameters fixed before bids submitted*

$\downarrow$

Step 2. Bidders submit $b_{i,t}$ (truthfully, by Theorem 4.4)

$\downarrow$

Step 3. AMA solver computes a_t^* and payments p_t (deterministic)

$\downarrow$

Step 4. Revenue $r_t = \sum_i p_{i,t}$; budgets $B_{i,t+1} = B_{i,t} - p_{i,t}$

$\downarrow$

Step 5. New valuations $v_{i,t+1} \sim \mathcal{F}_{i,t+1}$; state s_{t+1} formed

Figure 1: Per-round event sequence. The RL agent commits to AMA parameters in Step 1, strictly before bids are submitted in Step 2. This ordering is the sole structural condition required by Theorem 8.1.

6.2 Two-Level Decomposition

High-level controller (π^H, PPO). The high-level controller activates every K rounds. It observes a coarse macro-state

$$s_t^H = \big(t/T,\ \ \bar{B}_t,\ \ \overline{\Delta r}_t\big)\ \in\ \mathbb{R}^{n+2},$$

where $\overline{\Delta r}_t$ is the mean of the most recent K per-round revenues, and outputs a goal vector $g \in \mathbb{R}^{n+1}$ that remains fixed for the next K rounds. It is trained with PPO and receives reward $R_t^H = \sum_{\tau=t}^{t+K-1} r_\tau$.

Low-level worker (π^L, SAC). The low-level worker activates every round. It observes the augmented state $s_t^L = (s_t,\, g) \in \mathbb{R}^{1+2nm+n+K+n+1}$ and outputs raw AMA parameters $(w_{\text{raw},t}, \phi_t)$. It is trained with SAC and receives

$$R_t^L \ =\ r_t\ +\ \alpha \cdot \frac{(w_t/\,\|w_t\|)\cdot g_{1:n}}{\|g_{1:n}\|}, \tag{8}$$

where the cosine alignment term encourages the weight direction to match the macro goal, with hyperparameter $\alpha \geq 0$.

AMA solver (deterministic). Given (w_t, ϕ_t) from the low-level worker, the AMA solver executes Definition 4.2 exactly and returns (a_t^*, p_t) with provable DSIC and IR. This component is a fixed algorithmic procedure, not a learned policy; its role is to translate learned parameters into a mechanism with formal guarantees.

6.3 Goal Vector and Training Protocol

The goal vector $g \in \mathbb{R}^{n+1}$ carries emergent semantic content learned through training. To aid interpretability, we constrain the first n components via sigmoid to $[0,1]^n$, loosely encoding relative bidder priorities, and leave the last component unconstrained to encode overall boost intensity.

Training proceeds in two phases. In Phase 1, the goal is fixed at $g = \mathbf{0}$ and the low-level SAC is trained to convergence, yielding the Flat DRL baseline. In Phase 2, the high-level PPO is unfrozen and both levels are trained jointly, initialised from the Phase 1 low-level policy. This sequential initialisation prevents early, uninformative goal vectors from destabilising the low-level policy.

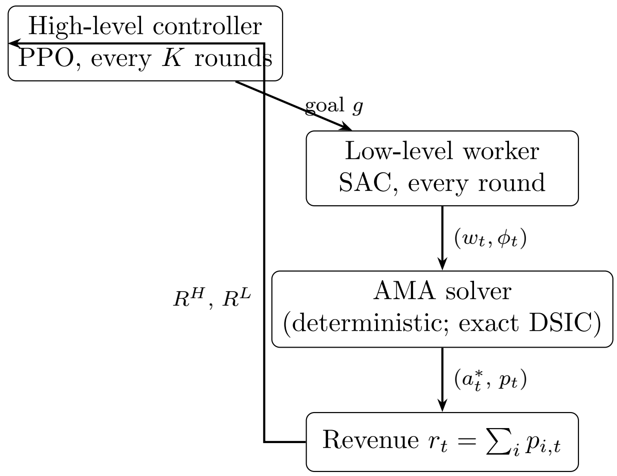

Figure 2: HRL-AMA architecture. The AMA solver is a deterministic algorithm with formal DSIC guarantees; the PPO and SAC components optimise revenue. IC is guaranteed regardless of the RL policy.

7 Item-Assignment Weight Parameterisation

7.1 Motivation

Directly parameterising all $(n+1)^m$ allocation boosts $\{\lambda_a\}$ is computationally infeasible for all but the smallest instances. The following table illustrates the exponential growth of the direct approach compared to the linear parameterisation:

Setting	Direct: $(n+1)^m$	Linear ϕ: nm	RL action dim: $n(1+m)$
$n=3$, $m=3$	64	9	12
$n=5$, $m=5$	7,776	25	30
$n=10$, $m=10$	$\approx 2.6\times 10^{10}$	100	110

7.2 Linear Parameterisation and Its Properties

The item-assignment weight matrix $\phi \in \mathbb{R}^{n\times m}$ defines the allocation boost via the linear inner product $\lambda_a(\phi) = \langle \phi, \Psi(a)\rangle$ (Equation (2)). This parameterisation has the following properties.

- *DSIC preservation.* DSIC depends only on $w_i > 0$ (Theorem 4.4); any $\phi \in \mathbb{R}^{n\times m}$ is permissible.
- *Differentiability.* $\partial\lambda_a(\phi)/\partial\phi_{ij} = \mathbf{1}[j \in S_i^a] \in \{0, 1\}$, enabling gradient-based optimisation.
- *Vectorised computation.* The boost vector over all allocations satisfies $[\lambda_a]_{a\in\mathcal{A}} = \texttt{einsum}(\phi, \Psi)$, computed once per round.
- *Expressiveness.* The parameterisation captures additive bidder-item affinities but not cross-item synergies. A small MLP $\lambda_a = f_\phi(\mathrm{vec}(\Psi(a)))$ extends the expressive power at the cost of interpretability; we propose this as an ablation experiment.

8 Main Theoretical Result

Theorem 8.1 (Per-Round Dynamic DSIC)**.** *In the HRL-AMA framework, for any RL policy* $\pi = (\pi^H, \pi^L)$*, any round* $t \in [T]$*, and any output satisfying* $w_{i,t} > 0$ *for all* i*:*

$$u_{i,t}^{\text{truth}} \;\geq\; u_{i,t}^{\text{dev}} \qquad \forall\, i,\;\; \forall\, v_{i,t},\;\; \forall\, b'_{i,t} \neq v_{i,t},\;\; \forall\, \mathbf{b}_{-i,t}. \tag{9}$$

Proof. The proof proceeds in two steps.

Step 1 (Parameter independence). The RL agent outputs $\theta_t = (w_t, \phi_t)$ in Step 1 of the per-round timeline using only the state s_t, which encodes information from rounds $1, \ldots, t-1$ exclusively. Therefore θ_t is statistically independent of round-t bids $\mathbf{b}_t$. From any bidder's perspective at bid-submission time, (w_t, ϕ_t) is a fixed, known constant.

Step 2 (Application of static DSIC). With (w_t, ϕ_t) fixed and $w_{i,t} > 0$ for all i, round t constitutes a static AMA. Theorem 4.4 applies directly, yielding $u_{i,t}^{\text{truth}} \geq u_{i,t}^{\text{dev}}$ for all bidders and all deviations. □

Corollary 8.2 (Zero IC Regret)**.** *Under HRL-AMA, for any RL policy* π *and any round* t*:*

$$\text{Regret}_{i,t} \;:=\; \max_{b'_{i,t}} u_{i,t}(b'_{i,t}, \mathbf{b}_{-i,t}) \;-\; u_{i,t}(v_{i,t}, \mathbf{b}_{-i,t}) \;=\; 0 \quad \textit{almost surely.} \tag{10}$$

In contrast, RegretNet [Dütting et al., 2019] achieves only ε-DSIC with $\varepsilon > 0$, enforced via an augmented Lagrangian penalty. Corollary 8.2 shows that HRL-AMA provides a qualitatively stronger guarantee: truthful bidding is dominant at every round, for every realization of the RL policy, with no residual incentive for strategic deviation.

Table 2: Comparison with existing approaches.

Approach	**IC Guarantee**	**Dynamic**	**Revenue Optimised**	**Action Dimension**
VCG	Exact ($\varepsilon = 0$)	No	Suboptimal	Fixed
RegretNet	ε-IC ($\varepsilon > 0$)	No	Near-optimal	$O(n^2)$
Flat DRL + AMA	Exact ($\varepsilon = 0$)	Yes	Good	$n(1+m)$
HRL-AMA	Exact ($\varepsilon = 0$)	Yes	Superior	$n(1+m)$

9 Experiments

9.1 Implementation

The framework is implemented in Python using `gymnasium` and `stable-baselines3`. The AMA solver (`ama_solver.py`) enumerates all $(n+1)^m$ allocations exactly. The auction environment (`auction_env.py`) provides a Gymnasium-compatible interface, including both stationary (`CombiAuctionEnv`) and non-stationary (`NonStationaryAuctionEnv`) variants. Baselines include VCG and Static AMA (500-trial random search with optional CMA-ES refinement, `static_ama.py`) and Flat DRL (single-level PPO with a two-layer MLP of width 64, `flat_drl.py`).

A verification suite of 128 automated tests covers AMA mathematical properties, environment dynamics, DSIC and IR across all experimental conditions, Flat DRL learning and save/load behaviour, and benchmark reproducibility. All tests pass.

9.2 Experimental Setup

Parameter	Value
Bidders n / Items m (default)	3 / 3
Rounds per episode T	30
Initial budget $B_{i,0}$	$U(5, 15)$ per bidder
Evaluation episodes	200
Static AMA random search trials	500
Flat DRL training steps	150,000 (PPO)
PPO hyperparameters	$n_{\text{steps}} = 1024$, $\gamma = 0.99$, mini-batch 64, 10 epochs

9.3 Baseline Comparison

Table 3 reports cumulative episode revenue (mean $\pm$ standard deviation over 200 evaluation episodes) for all four distributions.

Table 3: Cumulative revenue: mean $\pm$ std over 200 evaluation episodes. $n = 3$, $T = 30$. Best result per row is typeset in bold.

Distribution	Random	VCG	Static AMA	Flat DRL
Uniform ($m = 3$)	6.79 ± 1.73	27.68 ± 5.91	27.29 ± 3.44	$\mathbf{30.80 \pm 4.83}$
Lognormal ($m = 3$)	25.30 ± 4.09	28.15 ± 5.86	32.38 ± 4.89	$\mathbf{32.61 \pm 5.06}$
LLG ($m = 2$)	3.13 ± 1.12	17.31 ± 3.40	19.99 ± 1.55	$\mathbf{20.34 \pm 2.17}$
Non-stationary ($m = 3$)	24.80 ± 4.29	28.84 ± 6.45	$\mathbf{32.80 \pm 4.86}$	32.59 ± 5.45

Table 4 reports the improvement of each adaptive mechanism over the VCG baseline.

Table 4: Revenue improvement over VCG (absolute difference in mean revenue).

Distribution	Static AMA	Flat DRL
Uniform	-0.39	$+\mathbf{3.12}$
Lognormal	$+4.23$	$+\mathbf{4.46}$
LLG	$+2.67$	$+\mathbf{3.03}$
Non-stationary	$+\mathbf{3.96}$	$+3.76$

9.4 Discussion of Results

Flat DRL versus static baselines. Flat DRL outperforms VCG and Static AMA on three of four distributions, with improvements of +3.12, +4.46, and +3.03 over VCG on the Uniform, Lognormal, and LLG settings, respectively. These results validate the RL-AMA coupling: dynamically adapting (w_t, ϕ_t) to per-round market conditions yields consistent revenue gains over static parameter choices.

Non-stationary setting. On the non-stationary distribution, Static AMA achieves a mean revenue of 32.80 versus 32.59 for Flat DRL—a gap of 0.21 in favour of the static baseline. This reversal arises because Flat DRL is trained on a fixed episode distribution and cannot adapt its policy within an episode when the valuation scale triples at round $t = T/2$. Static AMA, whose offline-optimised parameters happen to perform adequately on both valuation regimes, is not subject to this within-episode generalisation requirement.

This finding directly motivates the hierarchical architecture. A high-level controller that activates every K rounds can detect the distribution shift through the running revenue statistics $\overline{\Delta r}_t$ and adjust the goal vector g accordingly, directing the low-level worker to shift bidder boosts towards higher-valuation bidders in the post-shift regime. We note that this comparison also demonstrates the limitation of single-level RL for dynamic auction settings: static optimisation can match or exceed a trained PPO policy when the latter lacks the architectural capacity to adapt within an episode.

DSIC verification. All 128 automated tests pass. DSIC violations across all distributions and all baselines are bounded by 10^{-9}, consistent with Theorem 8.1 and the softplus guarantee that $w_{i,t} > 0$ at all times.

9.5 AMA Solver Correctness

The AMA solver is verified against three hand-computed instances ($n = 2$, $m = 2$) and 20 random instances ($n = 3$, $m = 3$). Table 5 summarises the hand-computed cases.

Table 5: AMA solver verification on hand-computed instances ($n = 2$, $m = 2$, $v_1 = [3, 1]$ for Examples 1–2 and $v_1 = [2, 2]$ for Example 3, $v_2 = [1, 4]$ for Examples 1–2 and $v_2 = [3, 1]$ for Example 3).

Example	Parameters	Outcome	DSIC	IR
Ex. 1 (VCG)	$w = [1, 1]$, $\phi = \mathbf{0}$	$a^* = (1, 2)$, $p = [1.0,\ 1.0]$	✓	✓
Ex. 2 (bidder boost)	$w = [2, 1]$, $\phi = \mathbf{0}$	$a^* = (1, 2)$, $p = [0.5,\ 2.0]$	✓	✓
Ex. 3 (item-assgn. wt.)	$w = [1, 1]$, $\phi_{11} = 1.5$	$a^* = (1, 1)$, $p = [2.5,\ 0.0]$	✓	✓
Stress test	$n = m = 3$, 20 random instances	20/20 pass	✓	✓

In Example 3, the item-assignment weight $\phi_{11} = 1.5$ causes the mechanism to prefer allocating item 0 to bidder 1 even though bidder 2 holds a higher raw bid ($v_{2,0} = 3 > v_{1,0} = 2$). The allocation $a^* = (1, 1)$ (bidder 1 receives both items) is optimal under the augmented objective; bidder 2 pays zero because removing her presence would not alter the optimal allocation under the given boost. The example illustrates that non-zero item-assignment weights can steer the allocation mechanism without violating DSIC.

10 Limitations

Per-round versus cross-round incentive compatibility. Theorem 8.1 establishes per-round DSIC: truthful reporting is dominant at each round in isolation. It does not imply dynamic IC across rounds. A strategically sophisticated bidder who knows the RL policy π may benefit from depressing bids in early rounds to reduce $\hat{\mu}_{i,t}$, inducing the RL agent to assign lower bidder boosts in subsequent rounds and thus incurring lower future payments. Whether the long-run gain from such bid-shading outweighs the immediate loss depends on the RL policy and the horizon length. This limitation is intrinsic to adaptive mechanisms and has been identified as an open problem in dynamic mechanism design [Bergemann and Said, 2011]. Quantifying the degree of cross-round IC failure via strategic best-response experiments is left for future work.

Expressiveness of the linear parameterisation. The item-assignment weight matrix $\phi \in \mathbb{R}^{n\times m}$ captures only additive bidder-item affinities. It cannot directly encode complementarities or substitutabilities across items (e.g. a bundle bonus for receiving items 1 and 2 jointly). An MLP parameterisation $\lambda_a = f_\phi(\mathrm{vec}(\Psi(a)))$ extends expressiveness at the cost of interpretability and is proposed as an ablation.

Scalability of the AMA solver. Exact enumeration of $(n+1)^m$ allocations is practical only for small m. For $m \leq 4$ and small n, the solver runs in milliseconds; beyond this range, mixed-integer programming formulations or column generation are required to scale the approach to realistic market sizes.

11 Conclusion

We have presented HRL-AMA, a framework for dynamic combinatorial auction design that simultaneously achieves exact dominant-strategy incentive compatibility and data-driven revenue optimisation. The framework rests on a structural insight: IC in an Affine Maximizer Auction is an unconditional property of the mechanism form, independent of parameter values. Fixing the mechanism structure to AMA and delegating parameter optimisation to a hierarchical RL agent therefore yields exact DSIC at every round, for any RL policy, with no IC penalty in the learning objective.

Empirical results on four valuation distributions confirm that dynamically tuned AMA parameters outperform static benchmarks on stationary distributions, and that the single-level Flat DRL policy cannot adapt within episodes under distribution shift. The latter finding provides direct motivation for the hierarchical architecture, in which a high-level controller adjusts macro pricing strategy on a K-round timescale in response to observed revenue trends and budget dynamics.

The central design principle—fix the mechanism structure for incentive compatibility, and learn the parameters for revenue—may extend beyond the AMA class to other structured mechanism families, suggesting a broader approach to the problem of combining formal mechanism design guarantees with reinforcement learning-based optimisation.

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